

BIS REFERENCE HANDBOOK: ELECTRONICS, AI, IOT & SOFTWARE TESTING

EMERGING TECHNOLOGY STANDARDS (LITD COUNCIL PUBLICATIONS)

Formulated by the LITD 30 (Artificial Intelligence), LITD 27 (IoT), and LITD 15 (Software & Systems Engineering) committees, this handbook outlines national standards for digital trust.

1. ARTIFICIAL INTELLIGENCE STANDARDS

- **IS/ISO/IEC 22989 : 2022:** Information technology - Artificial Intelligence - Concepts and Terminology. Defines foundational terms (ML, Neural Networks, Bias, Explainability, Robustness).
- **IS/ISO/IEC 42001 : 2023:** Artificial Intelligence Management System (AIMS). International certifiable standard for responsible development, deployment, and risk management of AI systems.
- **IS/ISO/IEC 23894 : 2023:** Guidance on Risk Management for Artificial Intelligence.

2. INTERNET OF THINGS (IOT) & CYBERSECURITY

- **IS/ISO/IEC 30141 : 2018:** IoT Reference Architecture (Domain model, trust boundaries, gateway connectivity).
- **IS/ISO/IEC 27001 : 2022:** Information Security, Cybersecurity and Privacy Protection Management Systems.
- **IS 17428 : 2020:** Data Privacy Assurance - Engineering and Management requirements for personal data protection in India.

3. SOFTWARE QUALITY & SOFTWARE TESTING STANDARDS

Standard Number	Standard Title	Engineering Application Scope
IS/ISO/IEC 25010 : 2023	Systems and Software Quality Requirements and Evaluation (SQuaRE)	Defines 8 quality characteristics: Functional suitability, Performance, Compatibility, Usability, Reliability, Security, Maintainability, Portability
IS/ISO/IEC 29119 (Part 1-5)	Software Testing Standard Series	Part 1 (Concepts), Part 2 (Test Processes), Part 3 (Test Documentation - Test Plan, Test Case, Defect Report), Part 4 (Test Techniques - Boundary Value, Equivalence Partitioning), Part 5 (Keyword Testing)
IS/IEC 61508	Functional Safety of Electrical / Electronic / Programmable Safety Systems	Safety Integrity Levels (SIL 1 to SIL 4) for embedded software in railway, medical, and aerospace control systems